

Array multiplier

Unsigned multiply builds partial products from every bit pair

Five times three starter

- Starter: four-bit unsigned A equals five, B equals three
- Five is zero-one-zero-one, three is zero-zero-one-one
- Partial row for B-zero is five; row for B-one shifted left is ten
- Four AND cells are one in the grid
- Add the partials and the product is fifteen, binary zero-zero-one-one-one-one
- Product bit zero is A-zero AND B-zero only

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Array multiplier grid

See every partial product $A_i \cdot B_j$ on a shifted grid, then the unsigned product. Starter:
4-bit $5 \times 3 = 15$.

Starter example: 4-bit unsigned 5×3 — partial-product rows for B_0 and B_1 , product $001111_2 = 15$.

Challenge 0 / 22 cleared

Quiz: partial product: Each cell $A_i B_j$ in an unsigned array multiplier is...

☐ an AND of one multiplicand bit and one multiplier bit

☐ an XOR only

☐ a flop

☐ a $\lceil \log_2 \rceil$

Show hint

Check

Next

Idle

Quiz: partial product

Quiz: row shift

Quiz: product width

Quiz: $B_i=0$

Quiz: why "array"

5 × 3

7 × 9

15 × 15

× 0

× 1

PP ones for 5×3

3-bit 5×3

3-bit 7×7

Quiz: LSB product

Quiz: signed?

6 × 6

12 × 5

Partial row0

Partial row1

Quiz: summing

Quiz: area growth

8 × 4

Array multiplier starter

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Workbook practice

- In the workbook track, draw the AND grid for three-bit two times three
- Write partial row zero and row one with shifts
- Compute product bit zero by hand
- Explain why N-bit times N-bit needs two-N bits
- Name one pitfall: forgetting the left shift on each B-j row

Pitfalls to watch

- Do not confuse the AND grid with the final sum, adders still reduce the rows
- Signed multiply needs Booth or similar, not this unsigned array alone
- Area and delay scale with width; real chips use Wallace trees or Booth
- And remember: the browser lab is literacy
- Real RTL still needs pipelining and signed/unsigned rules

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, sketch one partial row and one shifted row
- When you are ready, take the short quiz, then continue to the ALU explorer